## Derivación Manual

$$R(p) = \log\left(\frac{K^A(p)}{K^V(p)}\right) \qquad K^A(p) = M + \sqrt{2F K^A(p)} \qquad K^V(p) = M + \sqrt{2F K^V(p)}$$

$$R(p) = \log K^A(p) - \log K^V(p)$$

$$R_p = \frac{\partial \log K^A(p)}{\partial p} - \frac{\partial \log K^V(p)}{\partial p}$$

$$= \frac{K_p^A(p)}{K^A(p)} - \frac{K_p^V(p)}{K^V(p)}$$

$$K_p^V = \frac{\partial}{\partial p}\left[M + \sqrt{2F K^V(p)}\right]$$

$$= \frac{\partial}{\partial p}\left(2F K^V(p)\right)^{1/2}$$

$$= \frac{1}{2}\left(2F K^V(p)\right)^{-1/2} \frac{\partial}{\partial p}\left(2F K^V(p)\right)$$

$$= \frac{\sqrt{2F}}{2\sqrt{K^V(p)}} K_p^V(p)$$

$$R_p = \frac{\frac{\sqrt{2F}}{2\sqrt{K^A}} K_p^A}{M + \sqrt{2F K^A}} - \frac{\frac{\sqrt{2F}}{2\sqrt{K^V}} K_p^V}{M + \sqrt{2F K^V}}$$

$$= \frac{\sqrt{2F}}{2}\left[\frac{K_p^A}{\sqrt{K^A}\left(M + \sqrt{2F K^A}\right)} - \frac{K_p^V}{\sqrt{K^V}\left(M + \sqrt{2F K^V}\right)}\right]$$

$F > 0 \qquad M \geq 0 \qquad K^A > 0 \qquad K^V > 0$

$\frac{\sqrt{2F}}{2} > 0 \qquad \sqrt{K^A}\left(M + \sqrt{2F K^A}\right) > 0 \qquad \sqrt{K^V}\left(M + \sqrt{2F K^V}\right) > 0$

$$R_p > 0 \iff \frac{K_p^A}{\sqrt{K^A}\left(M + \sqrt{2F K^A}\right)} > \frac{K_p^V}{\sqrt{K^V}\left(M + \sqrt{2F K^V}\right)}$$

$$K_p^A > 0 \not\Rightarrow R_p > 0$$

$$\Rightarrow R_p = \frac{\sqrt{2F}}{2} \frac{K_p^A}{\sqrt{K^A}\left(M + \sqrt{2F K^A}\right)}$$

$F > 0 \qquad M \geq 0 \qquad K^A > 0 \qquad K_p^A > 0$

$$\Rightarrow R_p > 0$$

$K_p^V = 0 \quad K_p^A > 0 \longrightarrow R_p > 0$

$K_p^A > 0$ por sí sola no determina el signo de $R_p$